

Scope of Works:

- 1. 24F PULL-BACK FOC INSTALLATION
- 2. FOC INSTALLATION
- 3. ODF, NAP & LCP INSTALL TION
- 4. FOC End to End Testing
- 5. Acceptance

Pullback Cable Installation Guidelines:

Cable Laying Procedure:

- 1. Unroll the cable by the pay-off stand release cable in figure 8 stroke on the floor.
- 2. Cut the cable to the required total length
- 3. Relieve the twist and set the cable as straight as possible. If the space is limited, setting the cable in U-Track is recommended.
- 4. Fix the fiber cores on the cable sheeth with adhesive tape at the top end of the cable.
- 5. Lay the cable from the top floor to the basement through the bus duct without any twist
- 6. Identify the maintenance loop for proper placement.
- 7. Fix the cable in bus duct with tie rope.
- 8. Fix the top end. Create loops and remove the adhesive tape to free the cable core.

Midspan Access

- 1. Create an extraction window where fiber core will be routed to the NAP.
- 2. Create a pick window in a higher level. Make sure the distance between the window will cover the length between extraction window and the user or core in the NAP.
- 3. Select the required fiber cores in the pick window and cut off.
- 4. Pull the selective fiber core back from the extraction window and lay them to the ODU or NAP.
- 5. For complex scenario, special protection are required on the set up to prevent fiber break, no trace and high loss.

Midspan Access

- 1. This cable should not be twisted during the whole installation.
- 2. For cable laying in vertical tube, pulling rope is recommended and tension should be observed not to exceed as what the specs prescribed.
- 3. Proper bending must be observe (100 during installation and 200 under operation).

PROPOSED FTTB PLAN OF FTTB SITE

ADDRESS: (Site Location with coordinates)

PCN: XXXXX
DISTRICT: XXXXXX
PROVINCE: XXXXX
CITY: XXXXX
BARANGAY: XXXXX

POI INFORMATION	
Homing Exchange	-
Feeder Cable ID	(Feeder Cable ID)
Core Assignment	-
Distance from POI to OLT	--m
Distance from POI to ODF	--m
OLT Code	-
OLT Name	-

LINK BUDGET LOSS

LINK BUDGET LOSS @ 1550nm 0.25db/km											
ODF Connector 0.3db/Each	ODF TO LCP 0.25db/KM	LCP Connector 0.3db/Each	LCP(L1:8) 10db/Each	Splice in LCP 0.05db/Each	LCP NAP Fiber loss 0.25db/KM	Splice in NAP 0.05db/Each	NAP(L1:8) 10db/Each	Active joint in NAP 0.5db/Each	TOTAL(db)		
1	0.3	0.000	0.0000	1	0.3	1	10	1	0.05	1	10
LINK BUDGET LOSS @ 1490nm 0.36db/km											
ODF Connector 0.3db/Each	ODF TO LCP 0.3db/KM	LCP Connector 0.3db/Each	LCP(L1:8) 10db/Each	Splice in LCP 0.05db/Each	LCP NAP Fiber loss 0.36db/KM	Splice in NAP 0.05db/Each	NAP(L1:8) 10db/Each	Active joint in NAP 0.5db/Each	TOTAL(db)		
1	0.3	0.000	0.0000	1	0.3	1	10	1	0.05	1	10
LINK BUDGET LOSS @ 1310nm 0.36db/km											
ODF Connector 0.3db/Each	ODF TO LCP 0.3db/KM	LCP Connector 0.3db/Each	LCP(L1:8) 10db/Each	Splice in LCP 0.05db/Each	LCP NAP Fiber loss 0.36db/KM	Splice in NAP 0.05db/Each	NAP(L1:8) 10db/Each	Active joint in NAP 0.5db/Each	TOTAL(db)		
1	0.3	0.000	0.0000	1	0.3	1	10	1	0.05	1	10

POLE#	REASON FOR INSTALLED POLE
POLE# (0.1:4)	N/A
POLE# (0.027m)	N/A

BOM

TOTAL CABLE LENGTH (m) - (24F FOC-UG)	0m
TOTAL CABLE LENGTH (m) - (24F FOC-TRADITIONAL)	233m
TOTAL CABLE LENGTH (m) - (FIBER DROP)	12,002m
SINGLE LCP (1:8)	3
SINGLE NAP (1:8)	15
DOUBLE NAP 2(1:8)	0
TOTAL NUMBER OF NAP	3
TOTAL NUMBER OF LCP	15
PROPOSED 24F ODF - WALL MOUNTED	1
PROPOSED FLEXIBLE METAL CONDUIT	1
PROPOSED PVC MOUNDING	1
PROPOSED 19" 4 Legged Open Bay Rack	-
PROPOSED STRAIGHT CONNECTOR, pcs	-
PROPOSED FDB (450X380X120mm)	-
PROPOSED FDB (328X259X94mm)	-
PROPOSED 2"ø IMC PIPE (m)	-
PROPOSED 2"ø FMC PIPE (m)	-
PROPOSED 1.5"ø PVC PIPE (m)	1
PROPOSED KNOCKOUT	-
PROPOSED PULLBOX	-
PROPOSED COUPLING	-
PROPOSED FULL THREADED ROD (pcs)	-
PROPOSED BACKBOARD	-
101.6mm x 101.6mm PVC CABLE TRAY	-
EXISTING NETWORK PROVIDER	-

CREATED LINES	-
TOTAL NUMBER OF UNITS	-
ENDORSED PORTS	-
AVERAGE UNITS PER FLOOR	-
TOTAL NUMBER OF FLOORS	-

SYMBOLS	
 Dome-Type Closure	 Single NAP
 Loop	 Double NAP
 Grounding System	 Four NAP
 24F Fiber Pull Back Cable	 LCP
 Fiber Enclosure	 Rack-mounted ODF
 Equipment Rack	 Rack-mounted ODF
 Cable Label	 Rack-mounted ODF
 Pipe (PVC)	 Hanger Loop
 Loop	 Fiber Label

SYMBOLS	
 Loop	 DIPTEL
 Grounding System	 Pole (Cable Cons.)
 Extended message wire	 Pole (Wooden)
 Metal ROD	 Electric Pole (Cable, Square)
 Underground ROD	 Electric Pole (Cable, Circular)
 Loop	 Electric Pole (Wooden)
 Grounding System	 Lumbar
 Extended message wire	 Cable
 Metal ROD	 Cable
 Underground ROD	 Cable

PHILIPPINES

FIXED ACCESS ENGINEERING

SHEET NO.

APPROVED BY:

CHECKED BY:

DRAWN BY:

CERTIFIED BY:

ACKNOWLEDGED BY:

01

12

PROPOSED 24F Pullback Cable and LCP /NAP INSTALLATION

XXXXXXXXXX
at
XXXXXXXXXX
AND
XXXXXXXXXX

DESIGN MANAGER

DESIGN ENGINEER

OSP DESIGN ENGINEER

RF NETWORK ENGINEER

SP NETWORK ENGINEER

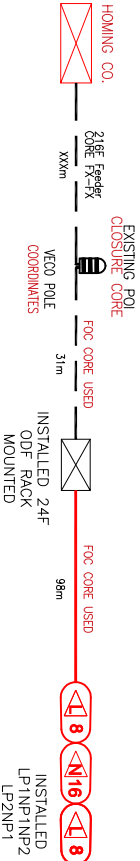
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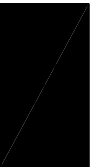
Scope of Works:

- 1. 24F PULL-BACK FOC INSTALLATION
- 2. FOC INSTALLATION
- 3. ODF LCP & NAP INSTALLATION
- 4. FOC End to End Testing
- 5. Acceptance

Site Location:

SINGLE LINE DIAGRAM



PROPOSED 24F Pullback Cable and LCP /NAP INSTALLATION at XXXXXXXXXXXX AND XXXXXXXXXXXX	PHILIPPINES				FIXED ACCESS ENGINEERING				SHEET NO. 02 12
	APPROVED BY:	CHECKED BY:	DRAWN BY:		CERTIFIED BY:		ACKNOWLEDGED BY:		
				DESIGN MANAGER				DESIGN ENGINEER	
				OSP DESIGN ENGINEER	AP. NETWORK ENGINEER			SP. NETWORK ENGINEER	TECHNICAL HEAD

BUILDING UNIT DETAILS			
FLOOR NO.	ACTIVE BLDG UNITS	INACTIVE BLDG UNITS	TOTAL BLDG UNITS
GRND FLOOR	8	-	8
2ND FLOOR	RETAILS	-	RETAILS
3rd FLOOR	PARKING	-	PARKING
5th FLOOR	PARKING	-	PARKING
6th FLOOR	PARKING	-	PARKING
7th FLOOR	OFFICE	-	OFFICE
8th FLOOR	OFFICE	-	OFFICE
9th FLOOR	OFFICE	-	OFFICE
10th FLOOR	AMENITIES	-	AMENITIES
11th FLOOR	17	-	17
12th FLOOR	17	-	17
14th FLOOR	17	-	17
15th FLOOR	17	-	17
16th FLOOR	17	-	17
17th FLOOR	17	-	17
18th FLOOR	17	-	17
19th FLOOR	17	-	17
20th FLOOR	17	-	17
21st FLOOR	17	-	17
22nd FLOOR	17	-	17
23rd FLOOR	17	-	17
24th FLOOR	17	-	17
25th FLOOR	2	-	2
26th FLOOR	2	-	2
TOTAL	233	-	233

 BUILDING UNIT DETAILS
N.T.S



 RISERDIAGRAM
N.T.S

PROPOSED 24F Pullback Cable and LCP /NAP INSTALLATION	at XXXXXXXXXXXX AND XXXXXXXXXXXX	PHILIPPINES				FIXED ACCESS ENGINEERING				SHEET NO. 03 12
		APPROVED BY:	CHECKED BY:	DRAWN BY:	CERTIFIED BY:	ACKNOWLEDGED BY:				
		DESIGN MANAGER	DESIGN ENGINEER	OSP DESIGN ENGINEER	RF NETWORK ENGINEER	SP NETWORK ENGINEER	TECHNICAL HEAD			

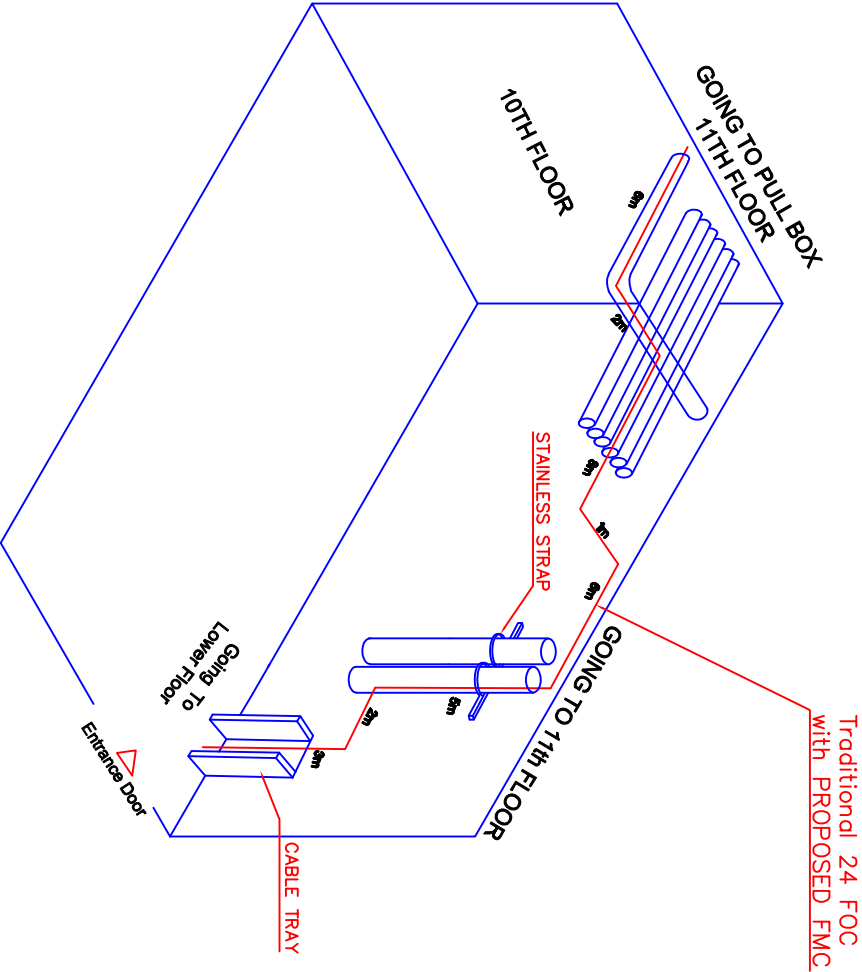
SUMMARY DETAILS FOR FTTB SITE						
ID	LATITUDE	LONGTTUDE	STATUS (NEW, EXISTING, FUTURE)	AVAILABLE PORTS (IF EXISTING ONLY)	DP ID	# OF PORTS
LP1	-	-	NEW	N/A		# OF PORTS
LP1NAP1	-	-	NEW	N/A		
LP1NAP2	-	-	NEW	N/A		
LP1NAP3	-	-	NEW	N/A		
LP1NAP5	-	-	NEW	N/A		
LP2	-	-	NEW	N/A		
LP2NAP1	-	-	NEW	N/A		
LP2NAP2	-	-	NEW	N/A		
LP2NAP3	-	-	NEW	N/A		
LP2NAP4	-	-	NEW	N/A		
LP2NAP5	-	-	NEW	N/A		

SUMMARY DETAILS FOR FTTB SITE						
ID	LATITUDE	LONGTTUDE	STATUS (NEW, EXISTING, FUTURE)	AVAILABLE PORTS (IF EXISTING ONLY)	DP ID	# OF PORTS
LP3	-	-	NEW	N/A		(# OF PORTS)
LP3NAP1	-	-	NEW	N/A		
LP3NAP2	-	-	NEW	N/A		
LP3NAP3	-	-	NEW	N/A		
LP3NAP4	-	-	NEW	N/A		
LP3NAP5	-	-	NEW	N/A		

SUMMARY DETAILS

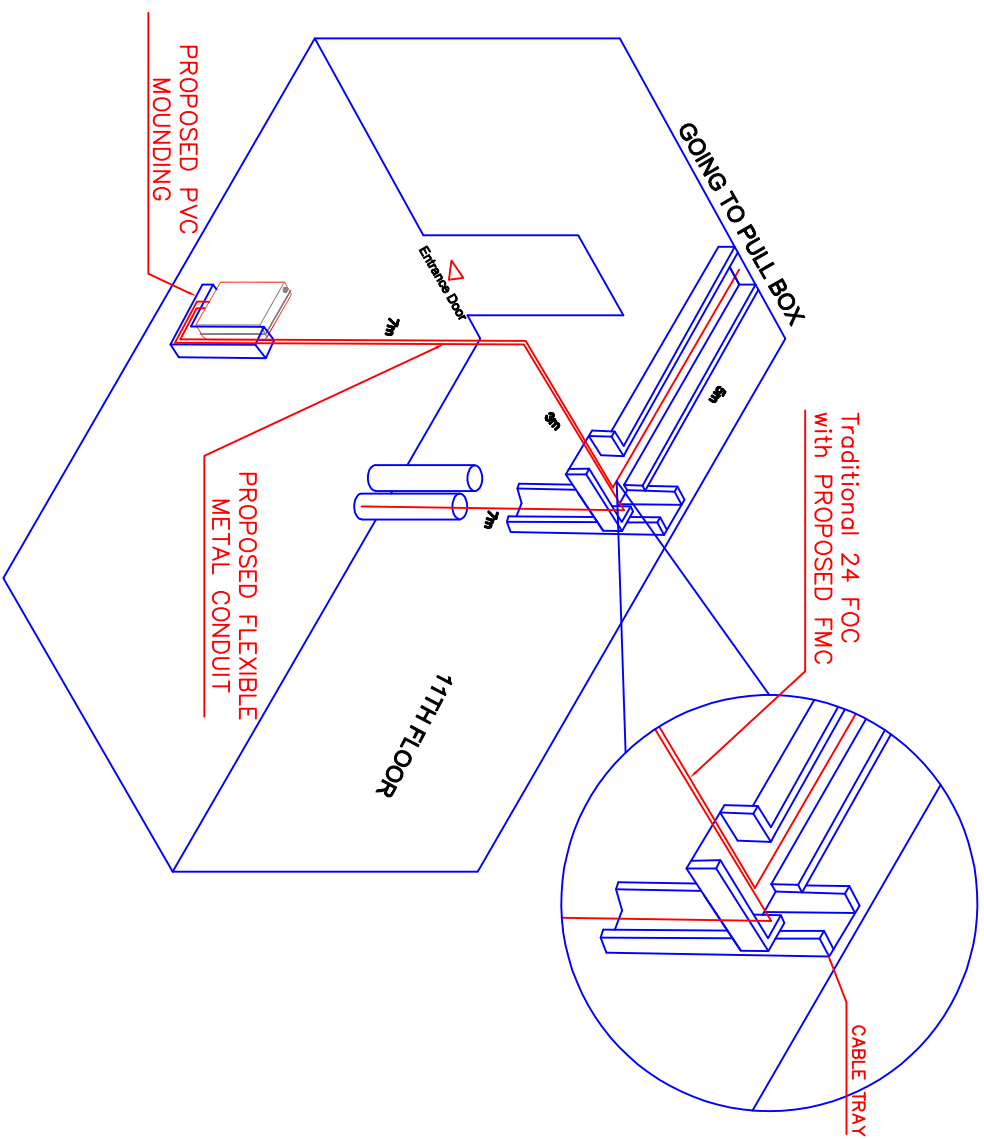
N.T.S

PROPOSED 24F Pullback Cable and LCP /NAP INSTALLATION at XXXXXXXXXXXX AND XXXXXXXXXXXX	PHILIPPINES		FIXED ACCESS ENGINEERING		SHEET NO. 04 12	
	APPROVED BY:	CHECKED BY:	DRAWN BY:	CERTIFIED BY:		
	DESIGN MANAGER	DESIGN ENGINEER	OSP DESIGN ENGINEER	AF NETWORK ENGINEER	SP NETWORK ENGINEER	TECHNICAL HEAD



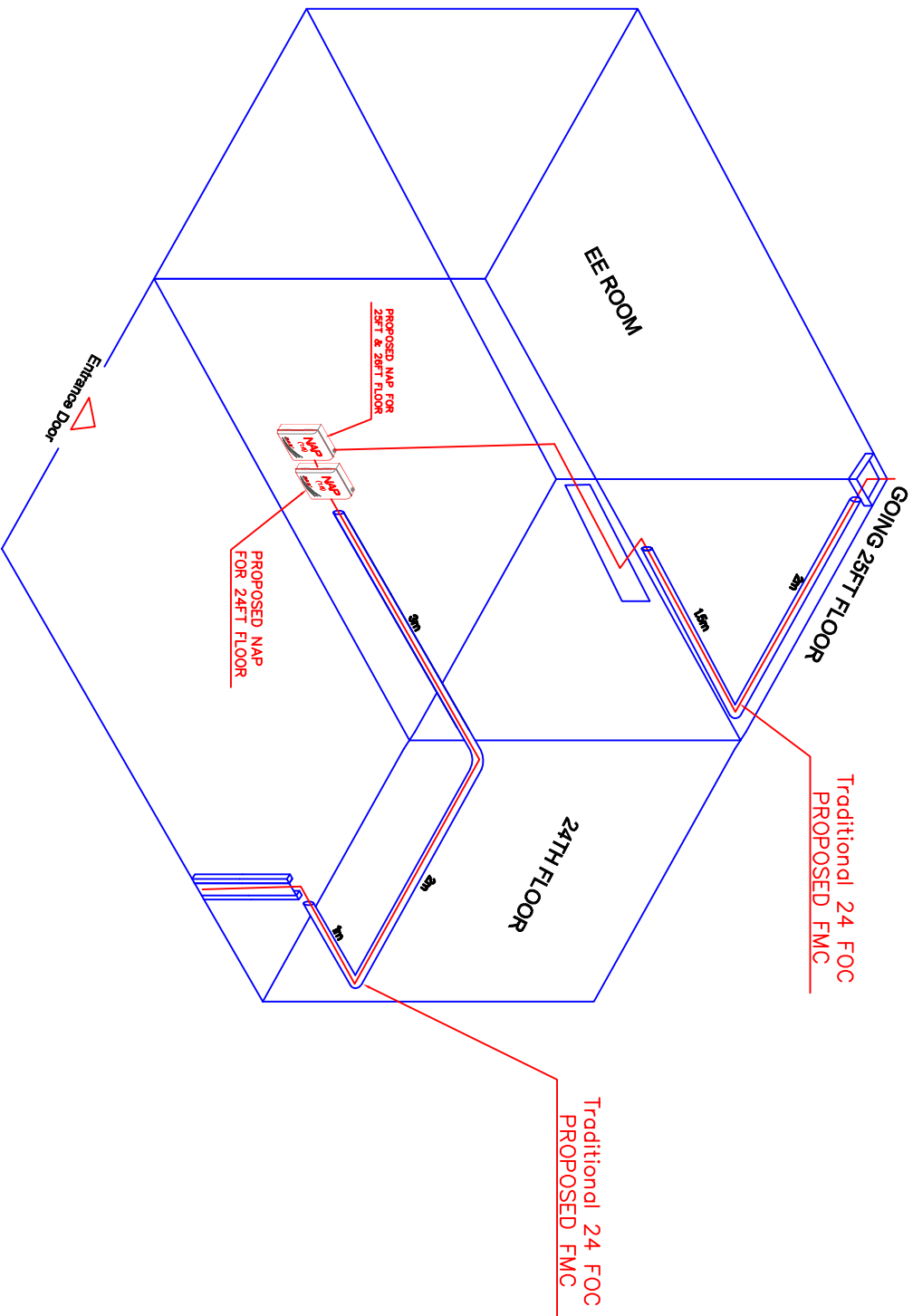
VERTICAL PLAN
ISOMETRIC VIEW
N.T.S

PROPOSED 24F Pullback Cable and LCP /NAP INSTALLATION at XXXXXXXXXXXX AND XXXXXXXXXXXX		PHILIPPINES				FIXED ACCESS ENGINEERING		SHEET NO. 06 12
		APPROVED BY:	CHECKED BY:	DRAWN BY:	CERTIFIED BY:		ACKNOWLEDGED BY:	
								
		DESIGN MANAGER	DESIGN ENGINEER	OSP DESIGN ENGINEER	SR NETWORK ENGINEER	TECHNICAL HEAD		



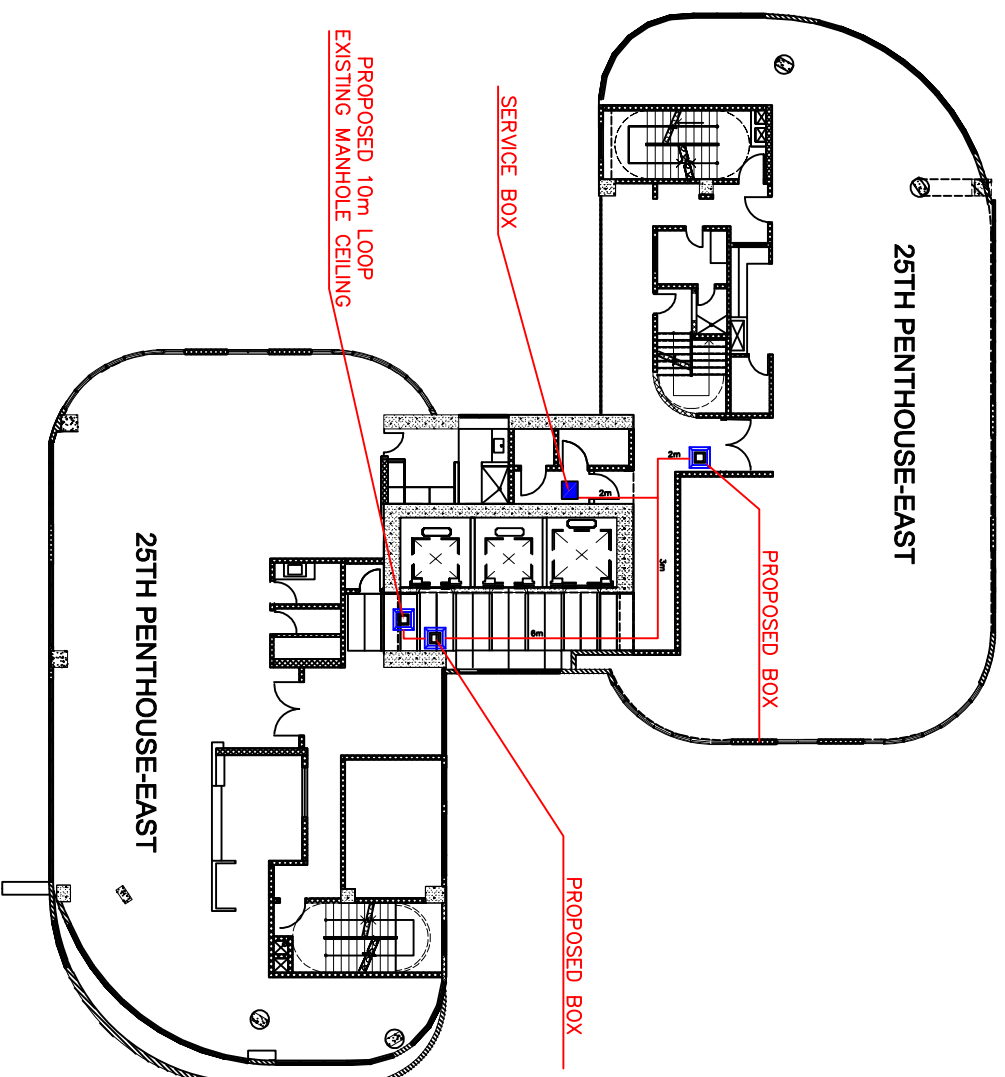
VERTICAL PLAN
ISOMETRIC VIEW
N.T.S

PROPOSED 24F Pullback Cable and LCP /NAP INSTALLATION at XXXXXXXXXXXX AND XXXXXXXXXXXX		PHILIPPINES		FIXED ACCESS ENGINEERING		SHEET NO.	
APPROVED BY:	CHECKED BY:	DRAWN BY:	CERTIFIED BY:	ACKNOWLEDGED BY:		07	
DESIGN MANAGER	DESIGN ENGINEER	OSP DESIGN ENGINEER	AS NETWORK ENGINEER	SP NETWORK ENGINEER	TECHNICAL HEAD	12	



VERTICAL PLAN
ISOMETRIC VIEW
N.T.S

PROPOSED 24F Pullback Cable and LCP /NAP INSTALLATION at XXXXXXXXXXXX AND XXXXXXXXXXXX		PHILIPPINES		FIXED ACCESS ENGINEERING		SHEET NO.	
APPROVED BY:	CHECKED BY:	DRAWN BY:	CERTIFIED BY:	ACKNOWLEDGED BY:		08	
DESIGN MANAGER	DESIGN ENGINEER	OSP DESIGN ENGINEER	SR NETWORK ENGINEER	TECHNICAL HEAD		12	



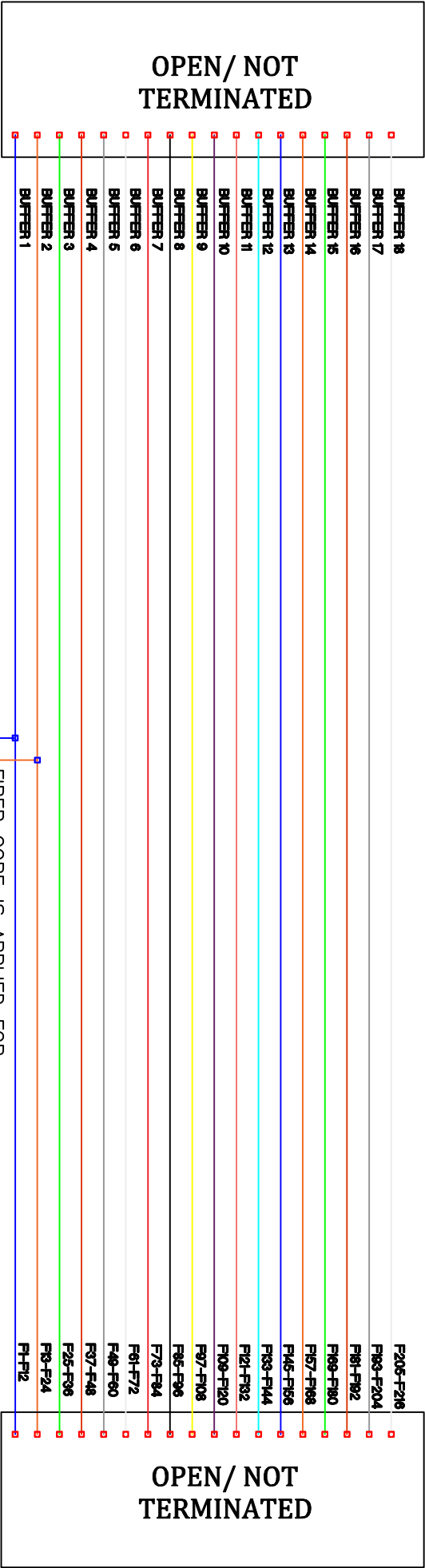
HORIZONTAL PLAN
LEVEL: 25F - 26F HALLWAY
N.T.S

PROPOSED 24F Pullback Cable and LCP /NAP INSTALLATION		PHILIPPINES		FIXED ACCESS ENGINEERING		SHEET NO.	
of XXXXXXXXXX AND XXXXXXXXXX	APPROVED BY:	CHECKED BY:	DRAWN BY:	CERTIFIED BY:		10 12	
							
	DESIGN MANAGER	DESIGN ENGINEER	OSP DESIGN ENGINEER	R. NETWORK ENGINEER	SR. NETWORK ENGINEER	TECHNICAL HEAD	

SPlicing DETAILS
FTTB SITE

HOMING CO.

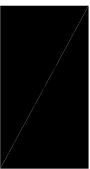
HOMING CO.

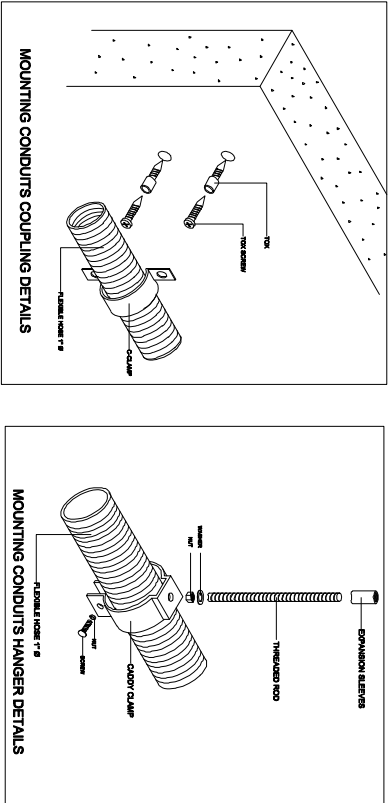
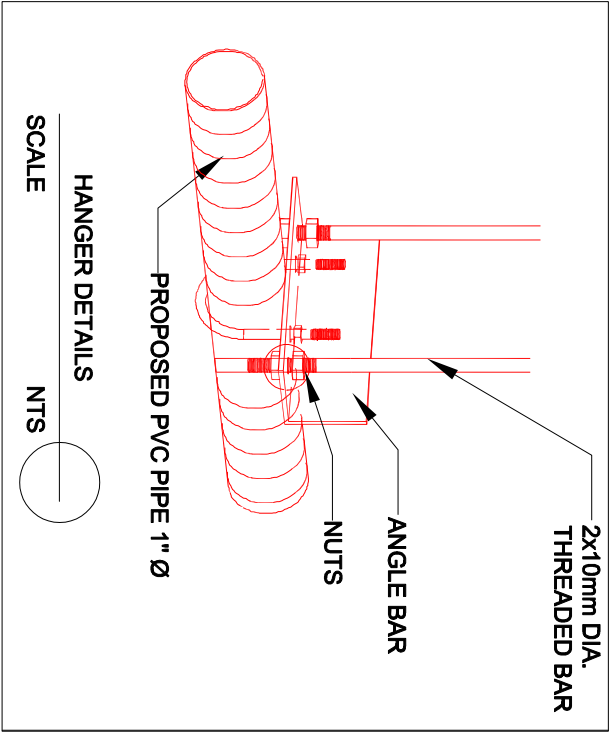
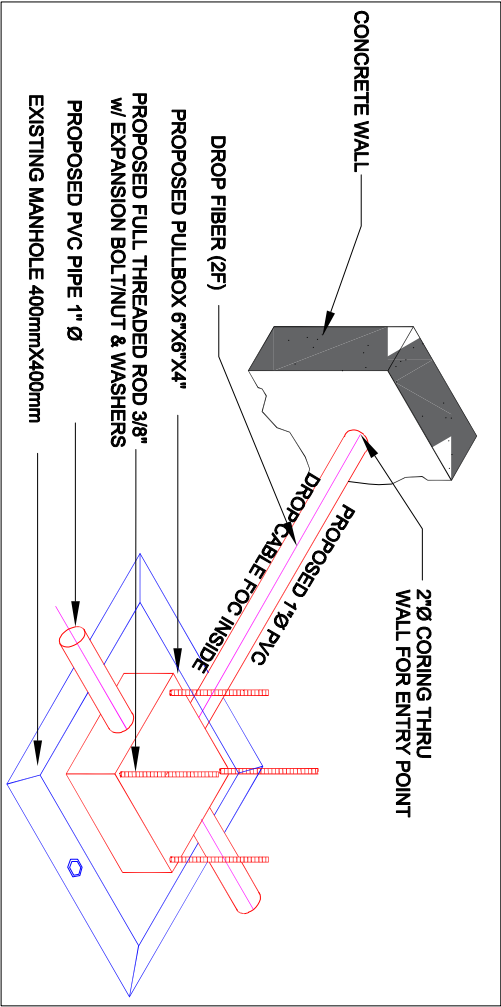
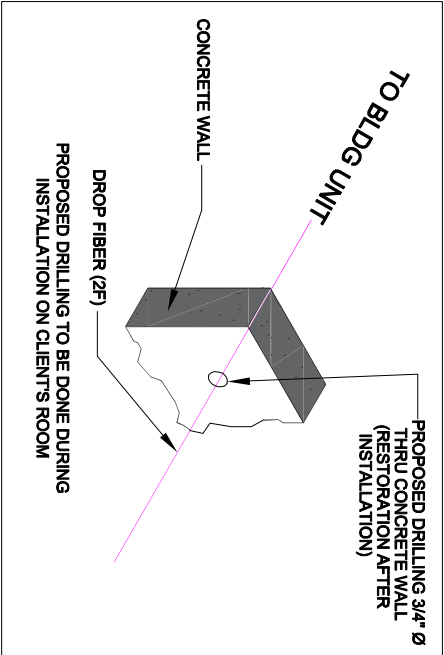


CABLE ID:
XXCABLEID

PROPOSED 24F ODF
SITE COORDINATES

SITE NAME

PROPOSED 24F Pullback Cable and LCP /NAP INSTALLATION		PHILIPPINES		FIXED ACCESS ENGINEERING		SHEET NO.	
at XXXXXXXXXXXX AND XXXXXXXXXXXX				APPROVED BY:		11	
		DESIGN MANAGER		CHECKED BY:		12	
		DESIGN ENGINEER		DRAWN BY:			
		OSP DESIGN ENGINEER		CERTIFIED BY:			
		SR NETWORK ENGINEER		ACKNOWLEDGED BY:			
		TECHNICAL HEAD					



PROPOSED 24F Pullback Cable and LCP /NAP INSTALLATION	PHILIPPINES				FIXED ACCESS ENGINEERING				SHEET NO.
	APPROVED BY:	CHECKED BY:	DRAWN BY:		CERTIFIED BY:		ACKNOWLEDGED BY:		
	XXXXXXX at AND XXXXXXX								
DESIGN MANAGER	DESIGN ENGINEER	OSP DESIGN ENGINEER	SR. NETWORK ENGINEER	TECHNICAL HEAD	12				12